SEEM5020: Final Project

Nguyen Trung Hieu
1155203613
1155203613@link.cuhk.edu.hk

Abstract

1 Algorithm Extension for strict Turnstile Model with α-Bounded Deletion property

1.1 Misra-Gries algorithm

The standard Misra-Gries algorithm is an insertion-only method, making the handling of deletions a primary challenge. To explicitly handle the Strict Turnstile Model with the α-Bounded Deletion property, the standard dictionary tracking is extended to process negative weights.The algorithm maintains a dictionary tracking up to k elements:Insertions ($v_t > 0$): If the element is tracked, its counter is incremented. If it is not tracked and the dictionary size is below capacity k, it is added. If the capacity is full, a collision occurs. The algorithm penalizes all currently tracked elements by decrementing their counters by the incoming weight. Any element whose counter drops to zero or below is explicitly evicted.Deletions ($v_t < 0$): Under the strict turnstile model, the true frequency of any individual element can never drop below zero ($f_e \geq 0$). If a negative update arrives for an actively tracked element, its counter is explicitly decremented. If this deletion causes the counter to reach zero or below, the element is completely removed from the dictionary. If a deletion arrives for an untracked element, it is ignored, operating under the assumption that it was previously evicted during a capacity collision.

1.2 Space-Saving algorithm

Space-Saving requires explicit structural extensions to handle deletions as a primary challenge. The extension maintains the core inheritance logic for insertions while introducing a strict purging mechanism for negative updates.Insertions ($v_t > 0$): If the item is tracked, its counter increments. If the capacity k is full and an untracked item arrives, the algorithm identifies the tracked item with the minimum counter. The minimum item is deleted, and the new item replaces it, inheriting the minimum counter's value plus the new insertion weight.Deletions ($v_t < 0$): If a negative update arrives for a tracked item, its counter is decremented. Under the strict turnstile model, the true frequency $f_e \geq 0$ at all times. Consequently, if an item's counter reaches zero or drops below it due to a deletion, it is permanently deleted from the dictionary to free up capacity.

1.3 Count-Min Sketch

While the standard Count-Min Sketch natively supports turnstile updates, it strictly overestimates true frequencies due to hash collisions. To optimize this under the α-Bounded Deletion property, the extension implements an Active Noise Cancellation mechanism. By tracking the exact total final weight of the stream (W), which is stabilized by the α-bound limit on total stream churn, the algorithm calculates the expected collision noise for any target bucket. The weight distributed among the other buckets in a given row is exactly $W -$ raw value. The expected noise is the average of

Preprint.

this remaining weight. This expected noise is subtracted from the raw bucket value to isolate the item's true frequency. To mitigate subtraction outliers, the algorithm takes the median of these noise-cancelled estimates across all depth rows, and finally clamps the result at zero to enforce the strict turnstile guarantee of $f_e \geq 0$.

1.4 Count Sketch

Count-Sketch natively supports turnstile updates and utilizes a secondary sign hash function ($+1$ or -1) to ensure the expected value of collision noise averages to zero. However, under tight memory constraints and high deletion churn, the variance within the buckets can swing wildly, resulting in extreme overestimations or negative frequency estimates. The extension implements a Variance Clipping filter explicitly designed for the Strict Turnstile Model. Because the total final weight of the stream (W) can be reliably tracked due to the α-Bounded Deletion property , and the strict turnstile model dictates that no frequency can drop below zero ($f_e \geq 0$), it is mathematically impossible for any single element's true frequency to exceed W. The algorithm bounds the raw median estimate by applying a strict mathematical clamp, ensuring the final estimate is $\max(0, \min(W, \text{raw_estimate}))$. This instantly corrects variance-induced outliers without requiring additional data structures.

2 Experimental Results and Analysis

2.1 Experimental Results

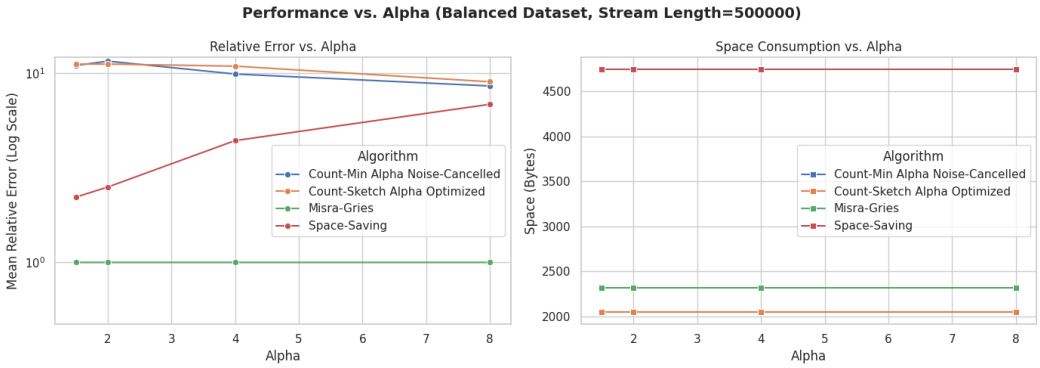

Figure 1: Balanced Performance vs Alpha

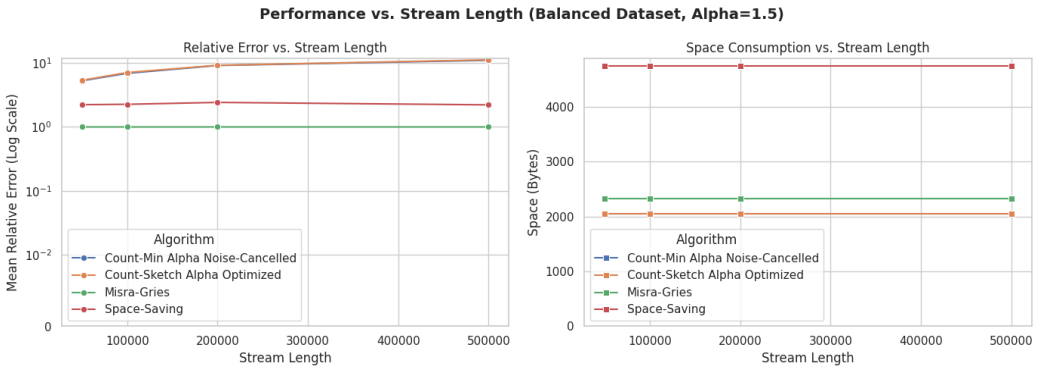

Figure 2: Balanced Performance vs Stream Length

2.2 Experimental Results Analysis

The following analysis details the performance of frequency estimation algorithms under restrictive deletion bounds (α).

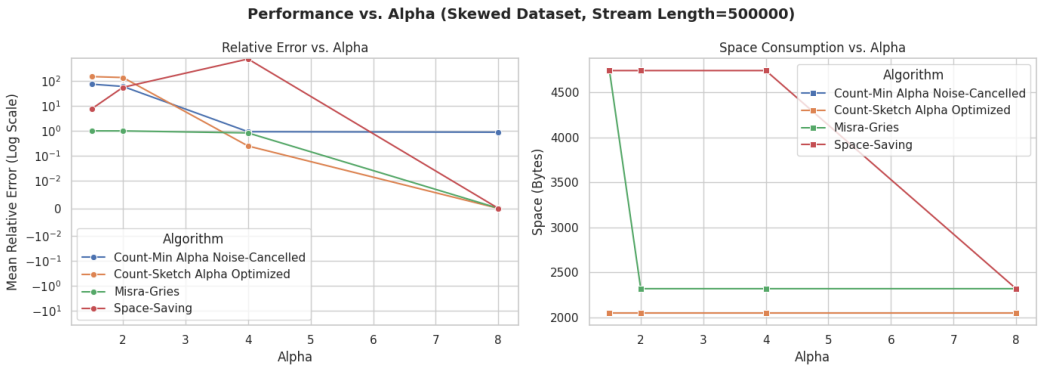

Figure 3: Skewed Performance vs Alpha

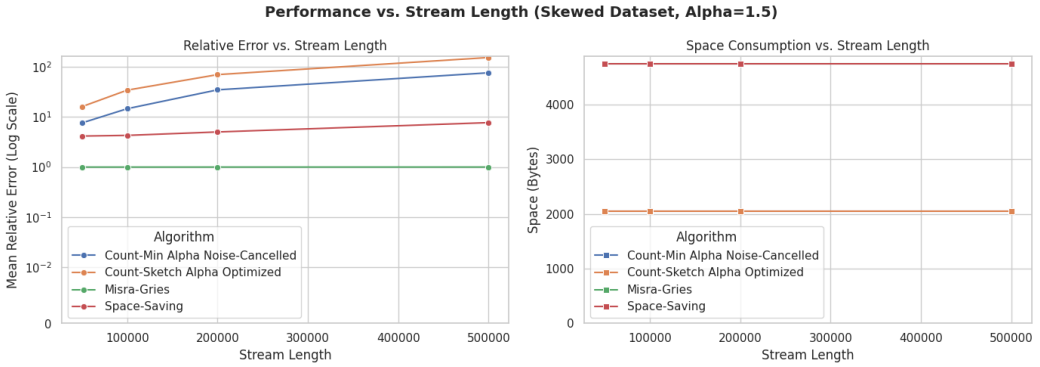

Figure 4: Skewed Performance vs Stream Length

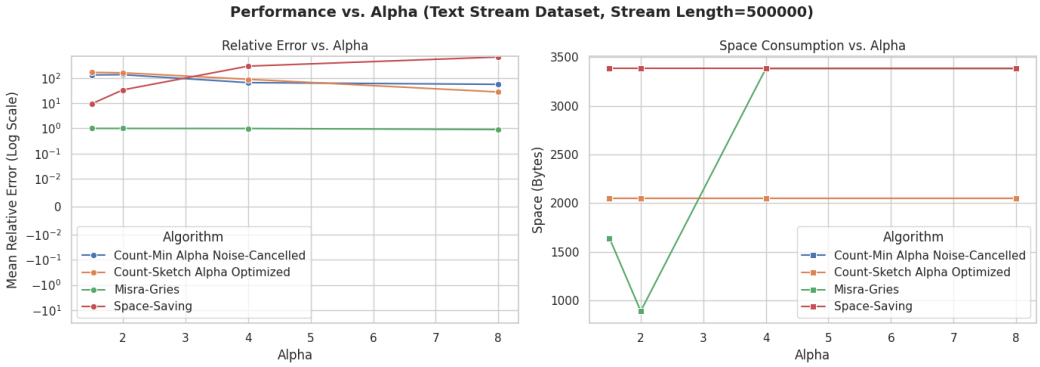

Figure 5: Text Stream Performance vs Alpha

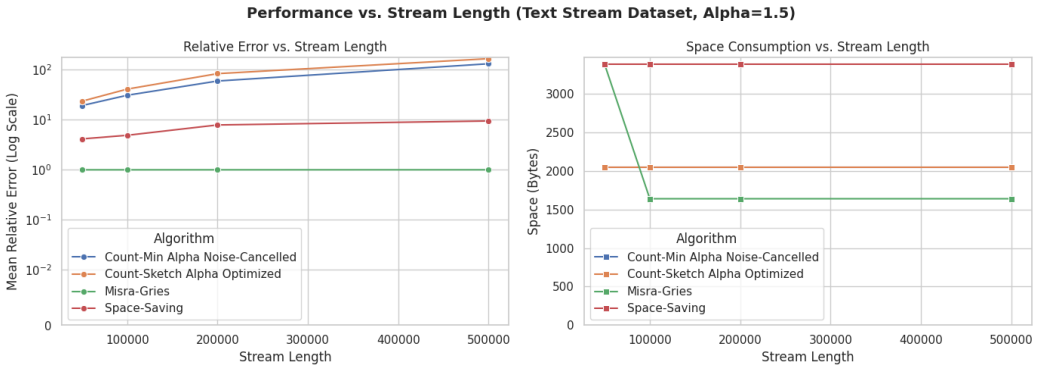

Figure 6: Text Stream Performance vs Stream Length

Space-Saving

- **Performance summary:** Dominates frequency estimation in skewed and real text streams under restrictive deletion bounds ($\alpha = 1.5$) by successfully anchoring to heavy hitters. However, it fails in balanced distributions and suffers catastrophic error spikes under high deletion churn.
- **Strengths:** Delivers the lowest error for Zipfian and natural text distributions when churn is low.
- **Weaknesses:** Consumes the most memory ($\approx 3400 - 4700$ bytes) due to dictionary and pointer overhead. Highly vulnerable to high α values, where new items inherit massively inflated residual errors from rapidly cycling elements.

Misra-Gries

- **Performance summary:** Functionally fails (100% amnesia, error of 1.0) across balanced datasets, real text streams, and most skewed configurations. The continuous arrival of long-tail or uniform items forces non-stop capacity collisions and uniform decrements, wiping the memory completely.
- **Strengths:** Moderate, dynamic memory footprint (≈ 2300 bytes). Achieves perfect accuracy (0.0 error) exclusively under extreme churn ($\alpha = 8$) on synthetic skewed data, where the long tail is mathematically eliminated.
- **Weaknesses:** Unusable in unpredictable or bursty text streams; the "double punishment" decrement logic aggressively destroys active heavy hitters before they can anchor.

Sketch Algorithms (Count-Min Alpha Noise-Cancelled & Count-Sketch Alpha Optimized)

- **Performance summary:** Both algorithms exhibit a steady, linear degradation in accuracy as stream length grows, driven by accumulating hash collisions. Unlike counter-based methods, they paradoxically improve at maximum deletion churn ($\alpha = 8$) on synthetic data because the active vocabulary drops below bucket capacity.
- **Strengths:** Most memory-efficient and perfectly static footprint (≈ 2050 bytes) regardless of data distribution or stream length. Immune to the catastrophic residual error compounding that breaks Space-Saving.
- **Weaknesses:** Cannot effectively isolate signal from noise in balanced (uniform) datasets. Maintains high relative error (10^1 to 10^2) on real text streams because natural vocabularies remain vast even under high churn.